Optical Flow and Motion Tracking Analysis

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Computer Vision Project

March 9, 2026

Abstract

This report presents an optical flow and motion tracking study on two video sequences, as required in the assignment. Two videos containing visible motion were selected: (i) a rotating Earth sequence and (ii) a natural scene containing a bird, rock, and flowing water. For each video, a 30-second segment was analyzed using dense optical flow computed with the Farnebäck method, and the results were visualized as videos, evidence frames, and motion statistics over time. In addition, motion tracking equations were derived from the brightness constancy assumption, a two-frame Lucas–Kanade formulation was established, and bilinear interpolation was described for subpixel intensity evaluation. Finally, two consecutive frames from each video were used to validate theoretical tracking predictions against actual pixel locations. The experimental results show that the rotating Earth video exhibits coherent global motion over the planet with a mostly static background, whereas the bird-and-water video exhibits stronger, more spatially irregular motion concentrated in the water region. Tracking validation achieved subpixel mean error in both cases, confirming that the theoretical tracking formulation is consistent with observed motion for the selected frames.

1 Introduction

Motion analysis is one of the core tasks in computer vision because temporal changes between frames reveal object displacement, scene dynamics, camera motion, and interactions between foreground and background. Optical flow estimates apparent pixel motion across consecutive frames, while motion tracking seeks to determine the displacement of selected points or image patches over time. Classical derivations of optical flow are commonly based on the brightness constancy assumption and local spatial smoothness, with foundational treatments given by Horn and Schunck and Lucas and Kanade [1, 2, 3].

The objective of this project is twofold. First, the project computes optical flow on two 30-second video samples and interprets what can be inferred from the flow fields. Second, it derives motion tracking equations from first principles, explains bilinear interpolation, and validates theoretical tracking results using actual pixel locations from consecutive video frames.

2 Video Data and Experimental Setup

Two input videos were used.

- **Video 1 (sample1.mp4):** rotating Earth against a dark background.
- **Video 2 (sample2.mp4):** bird on a rock with moving water in the scene.

The extracted metadata are summarized in Table 1.

Table 1: Video metadata for the two selected sequences.

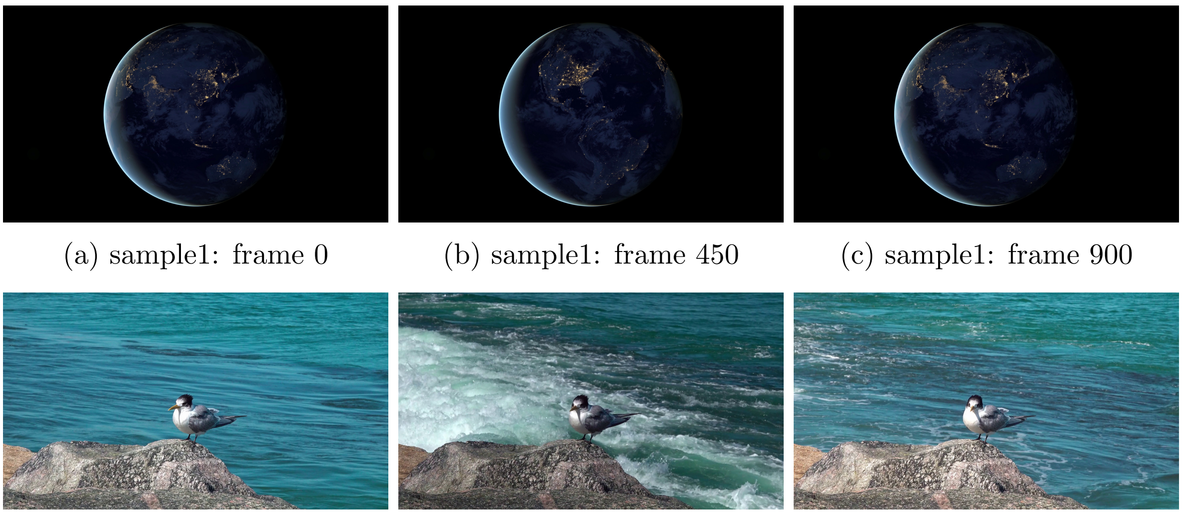
Video	FPS	Frames	Resolution	Duration (s)
sample1.mp4	30.0	901	1920 × 1080	30.0333
sample2.mp4	25.0	1068	1920 × 1080	42.72

For both videos, the first 30 seconds were used as the working segment. Three preview frames were extracted for each video to document the selected sequence. The frame indices used for preview extraction are listed in Table 2.

Table 2: Preview frames extracted from each 30-second segment.

Video	Preview 1	Preview 2	Preview 3
sample1.mp4	frame 0	frame 450	frame 900
sample2.mp4	frame 0	frame 375	frame 750

Figure 1 shows the extracted preview frames.



(a) sample1: frame 0 (b) sample1: frame 450 (c) sample1: frame 900

(d) sample2: frame 0 (e) sample2: frame 375 (f) sample2: frame 750

Figure 1: Preview frames from the two selected videos.

3 Part A: Optical Flow Computation and Visualization

3.1 Method

Dense optical flow was computed using the Farnebäck method, implemented in OpenCV through `calcOpticalFlowFarneback` [4, 5, 6]. This algorithm estimates a dense displacement field between consecutive frames by approximating image neighborhoods with polynomial expansions. In the implementation, each frame was resized to a working width of 960 pixels while preserving aspect ratio, converted to grayscale, and compared with the preceding frame.

The flow field was visualized in two complementary forms:

1. an HSV-based color visualization where hue encodes motion direction and value encodes motion magnitude, and

2. sparse arrow overlays sampled on a regular grid to show displacement vectors directly.

For each frame, the following statistics were also computed:

- mean optical flow magnitude,

- maximum optical flow magnitude, and

- moving-pixel ratio, defined here as the fraction of pixels with flow magnitude greater than 0.5.

The generated flow visualizations were exported as video files:

- `cv_project_output/sample1_optical_flow.mp4`

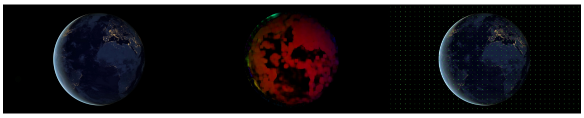
- `cv_project_output/sample2_optical_flow.mp4`

3.2 Optical Flow Results for sample1.mp4

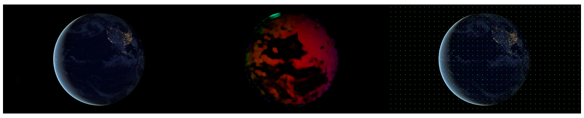
The summary statistics for the rotating Earth sequence are listed in Table. For `sample1.mp4`, the mean of the per-frame mean flow magnitude is 0.211598, the maximum observed flow magnitude is 5.672446, and the average moving-pixel ratio is 0.177897. The strongest mean motion occurs around frame 877, corresponding to 29.233 seconds, where the mean magnitude reaches 0.252801.

Qualitatively, the optical flow field is concentrated on the visible surface of the Earth, while the dark background remains near zero motion. This indicates that the method successfully separates the moving object from the static background. The flow on the planet is spatially coherent: neighboring regions move in related directions, which is consistent with the approximately rigid rotational motion of the globe as projected into the image plane. This sequence demonstrates that optical flow can reveal object boundaries, dominant direction of motion, and spatial variation in apparent velocity caused by projection and surface texture.

Representative evidence frames are shown in Figure 2, and temporal motion statistics are shown in Figure 3.

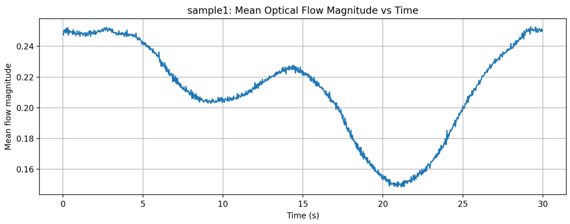


(a) Evidence frame near the first third of the clip.

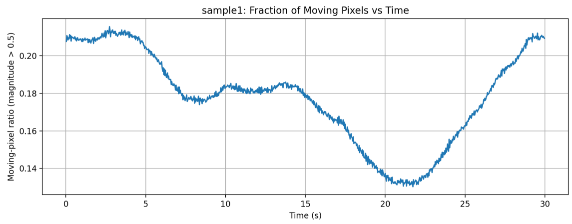


(b) Evidence frame near the second third of the clip.

Figure 2: Optical flow evidence frames for `sample1.mp4`. Each combined image contains the original frame, HSV flow visualization, and sparse arrow overlay.



(a) Mean flow magnitude versus time.



(b) Moving-pixel ratio versus time.

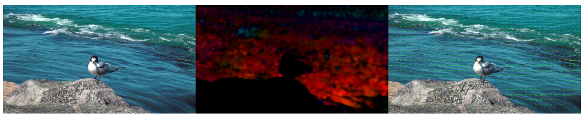
Figure 3: Temporal motion statistics for `sample1.mp4`.

3.3 Optical Flow Results for sample2.mp4

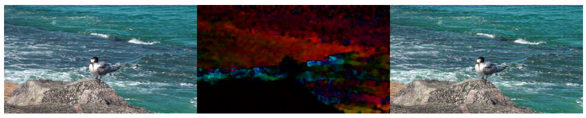
For `sample2.mp4`, the motion statistics are substantially larger than in the first video. The mean of the per-frame mean flow magnitude is 1.778561, the maximum observed flow magnitude is 38.318985, and the average moving-pixel ratio is 0.611745. The strongest mean motion occurs around frame 344 at 13.760 seconds, with mean magnitude 5.069956.

This result is consistent with the scene content. The bird and rock are comparatively static over the selected segment, while the flowing water produces strong, spatially varying, non-rigid motion. Unlike the Earth sequence, this video does not exhibit a single coherent global motion pattern. Instead, the water region contains multiple local motion directions and changing magnitudes over time. This demonstrates another important property of optical flow: it can distinguish rigid or nearly rigid motion from complex dynamic textures and can localize the areas where motion is concentrated.

Representative evidence frames are shown in Figure 4, and temporal motion statistics are shown in Figure 5.

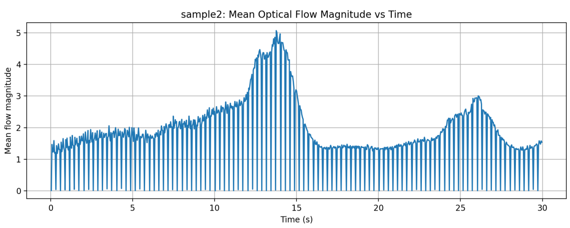


(a) Evidence frame near the first third of the clip.

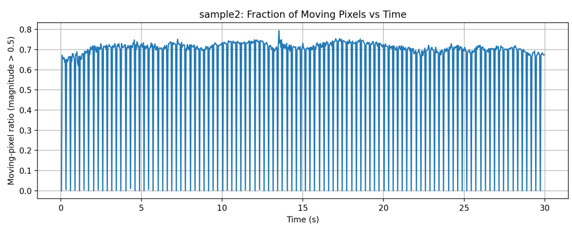


(b) Evidence frame near the second third of the clip.

Figure 4: Optical flow evidence frames for `sample2.mp4`. Each combined image contains the original frame, HSV flow visualization, and sparse arrow overlay.



(a) Mean flow magnitude versus time.



(b) Moving-pixel ratio versus time.

Figure 5: Temporal motion statistics for `sample2.mp4`.

3.4 Comparison Between the Two Videos

The two videos illustrate two different motion regimes. In `sample1.mp4`, motion is relatively weak in average magnitude and concentrated over the rotating Earth, producing a consistent global pattern. In `sample2.mp4`, motion is stronger, more irregular, and affects a much larger fraction of the image due to the flowing water. Numerically, the average moving-pixel ratio rises from 0.177897 in sample1 to 0.611745 in sample2, and the maximum magnitude rises from 5.672446 to 38.318985. Therefore, the second video contains both stronger motion and more widespread temporal variation.

4 Part B: Motion Tracking Theory

4.1 Brightness Constancy Assumption

Let $I(x, y, t)$ denote image intensity at spatial position (x, y) and time t. If a point moves by displacement (u, v) between two consecutive frames, the brightness constancy assumption states that its intensity remains approximately unchanged:

$$I(x, y, t) = I(x + u, y + v, t + 1). \tag{1}$$

This assumption is the starting point for a large family of optical flow and tracking methods [1, 2, 3].

4.2 Derivation of the Optical Flow Constraint Equation

Applying a first-order Taylor expansion to the right-hand side of (1) gives

$$I(x + u, y + v, t + 1) \approx I(x, y, t) + I_x u + I_y v + I_t, \tag{2}$$

where I_x, I_y, and I_t are the partial derivatives of intensity with respect to x, y, and t. Substituting into (1) yields

$$I(x, y, t) = I(x, y, t) + I_x u + I_y v + I_t. \tag{3}$$

Hence,

$$I_x u + I_y v + I_t = 0. \tag{4}$$

Equation (4) is the optical flow constraint equation.

4.3 Tracking Formulation for Two Frames

Equation (4) provides only one linear equation for the two unknown motion components u and v, so the motion at a single pixel cannot be determined uniquely. This is the classical aperture problem. Lucas and Kanade address this by assuming that all pixels in a small local window W undergo approximately the same motion [2]. For each pixel p_i inside the window,

$$I_x(p_i)u + I_y(p_i)v + I_t(p_i) = 0, \qquad i = 1, 2, \ldots, n. \tag{5}$$

These equations can be stacked into matrix form:

$$A \begin{bmatrix} u \\ v \end{bmatrix} = b, \tag{6}$$

where

$$A = \begin{bmatrix} I_x(p_1) & I_y(p_1) \\ I_x(p_2) & I_y(p_2) \\ \vdots & \vdots \\ I_x(p_n) & I_y(p_n) \end{bmatrix}, \qquad b = -\begin{bmatrix} I_t(p_1) \\ I_t(p_2) \\ \vdots \\ I_t(p_n) \end{bmatrix}. \tag{7}$$

The least-squares solution is

$$\begin{bmatrix} u \\ v \end{bmatrix} = (A^T A)^{-1} A^T b. \tag{8}$$

This forms the basis of local two-frame tracking. In practice, iterative refinement and image pyramids are often used to improve robustness for larger displacements [5, 6, 3].

4.4 Bilinear Interpolation

Tracked points often move to non-integer pixel locations. Therefore, image intensity and gradient values must be sampled at subpixel coordinates. Bilinear interpolation estimates the intensity at a continuous location (x', y') using the four nearest grid points. Let

$$x' = x_0 + \alpha, \qquad y' = y_0 + \beta, \tag{9}$$

where $x_0 = \lfloor x' \rfloor$, $y_0 = \lfloor y' \rfloor$, and $0 \le \alpha, \beta < 1$. Then

$$I(x', y') = (1-\alpha)(1-\beta)I(x_0, y_0) + \alpha(1-\beta)I(x_0+1, y_0) \tag{10}$$

$$+ (1-\alpha)\beta I(x_0, y_0+1) + \alpha\beta I(x_0+1, y_0+1). \tag{11}$$

Geometric image transformations in OpenCV use reverse mapping with interpolation to sample source images at non-integer coordinates, and linear interpolation is the default method for resizing and many related transformations [7, 8]. In this project, bilinear interpolation was implemented explicitly during iterative point tracking.

5 Part B: Tracking Validation on Consecutive Frames

5.1 Validation Procedure

To validate the theoretical tracking result, two consecutive frames were selected from each video. Salient points were detected in the first frame using a corner-based selection strategy. Their positions in the second frame were then predicted using an iterative Lucas–Kanade-style tracker based on Equation (8). Bilinear interpolation was used to evaluate intensities and gradients at subpixel positions. To obtain an empirical reference location, local exhaustive template matching was performed in the second frame. The predicted location and the best-matching actual location were compared using Euclidean distance in pixels.

5.2 Validation Results for sample1.mp4

For `sample1.mp4`, frames 150 and 151 were selected. The validation results are shown in Table 3. The mean tracking error is 0.382027 pixels and the maximum error is 0.548023 pixels. These small errors indicate that the theoretical local tracking model is highly consistent with the actual frame-to-frame displacement in the selected Earth sequence.

Table 3: Tracking validation results for `sample1.mp4` using frames 150 and 151.

Point	x_0	y_0	x_{pred}	y_{pred}	x_{act}	y_{act}	**Error**
1	500.0	221.0	501.4634	221.0399	501.0	221.0	0.4651
2	423.0	199.0	424.3760	199.1626	424.0	199.0	0.4097
3	472.0	220.0	473.4576	220.0781	474.0	220.0	0.5480
4	499.0	195.0	500.4005	194.9882	500.0	195.0	0.4007
5	472.0	192.0	473.4012	192.0552	473.0	192.0	0.4049
6	410.0	170.0	411.2553	170.1575	411.0	170.0	0.3000
7	511.0	138.0	512.1391	138.0040	512.0	138.0	0.1391
8	679.0	325.0	679.7366	324.7141	680.0	325.0	0.3887

Figure 6 shows the validation image. Points in the first frame are marked, while predicted and actual positions in the second frame are overlaid for visual comparison.

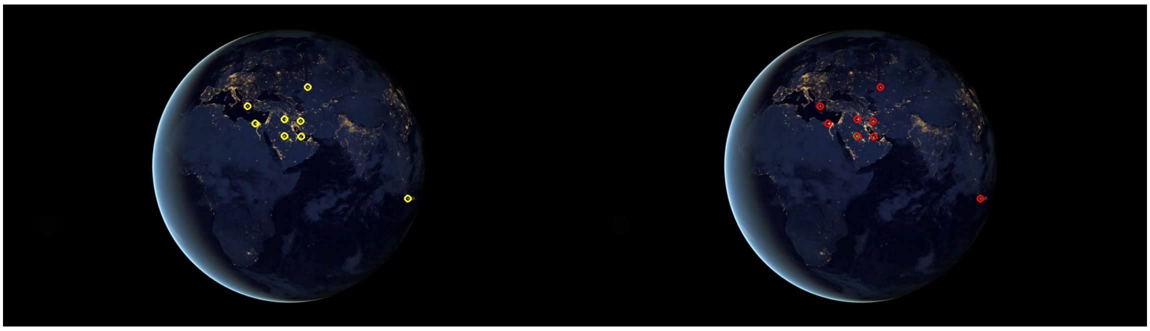


Figure 6: Tracking validation visualization for `sample1.mp4`. Predicted positions and matched actual positions are nearly coincident, which agrees with the low error values in Table 3.

5.3 Validation Results for sample2.mp4

For `sample2.mp4`, frames 186 and 187 were selected, and points were chosen in a region of interest containing stronger water motion. The validation results are shown in Table 4. The mean tracking error is 0.438999 pixels and the maximum error is 0.648365 pixels. The errors remain below one pixel, showing that the theoretical model still predicts motion accurately, although the sequence is more dynamic and locally irregular than the Earth sequence.

Table 4: Tracking validation results for `sample2.mp4` using frames 186 and 187.

Point	x_0	y_0	x_{pred}	y_{pred}	x_{act}	y_{act}	**Error**
1	550.0	149.0	546.6020	150.1842	547.0	150.0	0.4385
2	440.0	269.0	440.4835	268.9206	440.0	269.0	0.4900
3	846.0	168.0	847.4028	168.3710	847.0	168.0	0.5476
4	814.0	230.0	816.7655	230.1144	817.0	230.0	0.2610
5	525.0	296.0	524.9806	295.9099	525.0	296.0	0.0921
6	773.0	169.0	773.5116	169.3967	774.0	169.0	0.6292
7	483.0	307.0	483.6336	306.8623	483.0	307.0	0.6484
8	470.0	260.0	470.4051	260.0096	470.0	260.0	0.4052

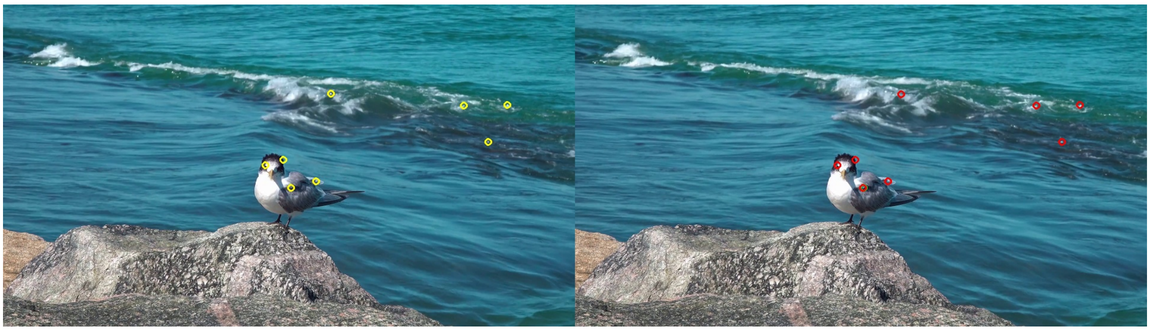


Figure 7: Tracking validation visualization for `sample2.mp4`. Predicted and actual locations remain close despite stronger and less regular local motion.

6 Discussion

The experiments clearly show how optical flow behavior depends on scene structure and motion type. The rotating Earth video produces low-to-moderate motion magnitudes, a mostly static background, and a coherent flow field concentrated on a single moving object. This is a favorable scenario for local tracking because the motion between consecutive frames is smooth and spatially structured.

The bird-and-water video is more challenging. The moving water produces larger magnitudes, a substantially higher moving-pixel ratio, and non-rigid, locally varying motion. Nevertheless, the frame-to-frame validation still achieved subpixel mean error. This suggests that the chosen window-based tracker remains effective when points are selected in textured regions and the inter-frame displacement is not too large.

Several limitations should also be noted. First, brightness constancy may be violated by illumination changes, specular reflections, or water shimmer. Second, optical flow estimation can fail in low-texture regions due to the aperture problem. Third, larger displacements may require multi-scale pyramids or more robust matching strategies. Finally, occlusion and motion blur can degrade both dense flow estimation and feature tracking [1, 2, 3, 5].

7 Conclusion

This project completed all required components of the assignment using two self-selected videos containing motion. For each 30-second sample, dense optical flow was computed, visualized as video, and analyzed quantitatively and qualitatively. The first video showed coherent global motion over the rotating Earth with a largely static background, while the second video showed stronger and more irregular motion associated with flowing water. The motion tracking equations were derived from the brightness constancy assumption, a Lucas–Kanade least-squares formulation was established, and bilinear interpolation was described for subpixel sampling. Finally, theoretical tracking predictions were validated on consecutive frames using actual pixel locations, resulting in mean errors of 0.382027 pixels for `sample1.mp4` and 0.438999 pixels for `sample2.mp4`. These results confirm that the theoretical formulation provides an accurate description of local inter-frame motion for the selected examples.

Supplementary Material

The following generated files accompany this report:

- Optical flow videos: `sample1_optical_flow.mp4`, `sample2_optical_flow.mp4`
- Flow statistics: `sample1_flow_stats.csv`, `sample2_flow_stats.csv`
- Evidence frames and plots in `cv_project_output/`
- Tracking validation tables and images for both videos
- Github Link: `https://github.com/xanyl/Optical-Flow-Analysis-and-Motion-Tracking.git`

References

[1] B. K. P. Horn and B. G. Schunck, "Determining optical flow," *Artificial Intelligence*, vol. 17, no. 1–3, pp. 185–203, 1981.

[2] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *Proceedings of the 7th International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.

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